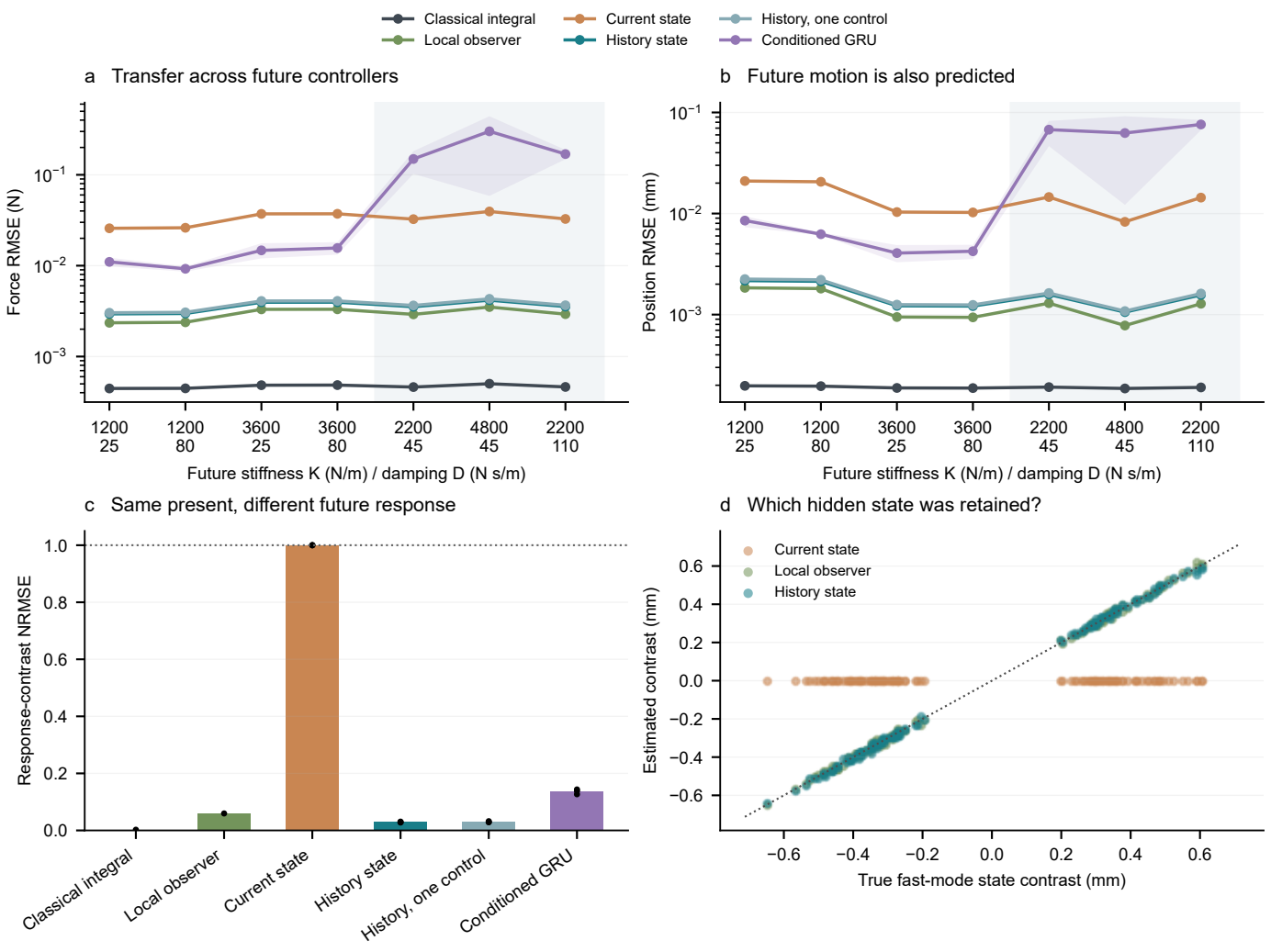

Classical integral
Local observer
Current state
History state
History, one control
Conditioned GRU
a Transfer across future controllers
Force RMSE (N)
10^{-1}
10^{-2}
10^{-3}
1200 25
1200 80
3600 25
3600 80
2200 45
4800 45
2200 110
Future stiffness K (N/m) / damping D (N s/m)
b Future motion is also predicted
Position RMSE (mm)
10^{-1}
10^{-2}
10^{-3}
1200 25
1200 80
3600 25
3600 80
2200 45
4800 45
2200 110
Future stiffness K (N/m) / damping D (N s/m)
c Same present, different future response
Response-contrast NRMSE
1.0
0.8
0.6
0.4
0.2
0.0
Classical integral
Local observer
Current state
History state
History, one control
Conditioned GRU
d Which hidden state was retained?
Current state
Local observer
History state
Estimated contrast (mm)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
True fast-mode state contrast (mm)